

\$330K Tax Reversal Automation

Tax Automation · Financial Recovery · Process Engineering

Executive Overview

Identified and remediated over \$330,000 in erroneous tax charges by designing and executing an automated tax reversal solution. The initiative reduced manual effort, minimized risk, and influenced future tax automation initiatives leveraging Vertex Batch Client Interface (BCI) and Alteryx.

Problem

In 2017, my employer had accumulated more than \$330,000 in incorrect tax assessments due to systemic posting errors across historical transactions. Existing processes relied on manual identification and reversal, which were time-intensive, error-prone, and not scalable. This created ongoing financial leakage and elevated audit and compliance risk. The Tax Technology group is tapped to resolve the issue before year end.

Approach

- Designed and led an automated tax reversal solution using Vertex O Series Batch Client Interface
- Programmatically identified invalid tax entries across historical transactions
- Executed bulk reversals with validation controls to ensure accuracy and auditability

Why This Matters

This case study demonstrates how targeted tax automation can directly drive financial recovery while reducing operational risk. It highlights the ability to translate tax domain knowledge into scalable, technology-enabled solutions with enterprise impact.

Transferable Skills

- Enterprise tax automation and system integration
- Financial recovery and error remediation
- Risk, controls, and audit awareness
- Cross-functional influence and stakeholder communication

Appendix: 1 - Automated Tax Test Case Generator — Lodging & Hospitality (Alteryx Designer)

Years later while working for another company, we needed a way to test content fast.

Challenge: Manually building Batch Client Interface (BCI) test files for Vertex's tax calculation engine was time-intensive and error-prone — especially for lodging transactions that span thousands of US jurisdictions, each with unique taxability rules, qualifying conditions, and flexible field requirements.

Solution: Designed and built an end-to-end Alteryx Designer workflow that **automatically generates production-ready BCI test files** for lodging/hospitality tax scenarios. The workflow:

- **Ingests** lodging transaction data, jurisdiction-level Tax Area references, and taxability mapping tables from multiple Excel and CSV sources.
- **Standardizes** Tax Area IDs and filters rules to isolate active, relevant taxability drivers (e.g., Flex-type drivers for the "Supplies" business event).
- **Enriches** each record by joining jurisdiction IDs, mapping qualifying conditions (up to 4 tiers with weighted comparison logic for thresholds like "Lodging Rooms"), and resolving taxability driver codes to product codes.
- **Assembles** fully formatted BCI transaction records — including dynamically constructed document numbers, date conversions, default units of measure, and result-type translations (Taxable, Exempt, etc.).
- **Outputs** a single, submission-ready CSV file conforming to Vertex's BCI specification.

Impact:

- ⚡ **Eliminated hours of manual test file preparation** per testing cycle.
- 🎯 **Improved accuracy** by programmatically enforcing field formatting, jurisdiction matching, and qualifying condition logic.
- 📊 **Scaled coverage** to thousands of US jurisdictions with consistent, repeatable test cases — supporting Vertex's content testing and validation initiatives.

Tools & Technologies: Alteryx Designer · Excel/CSV I/O · Regex Parsing · Multi-Tier Conditional Logic · Vertex BCI Specification

Title: Batch Client Interface Test Generator

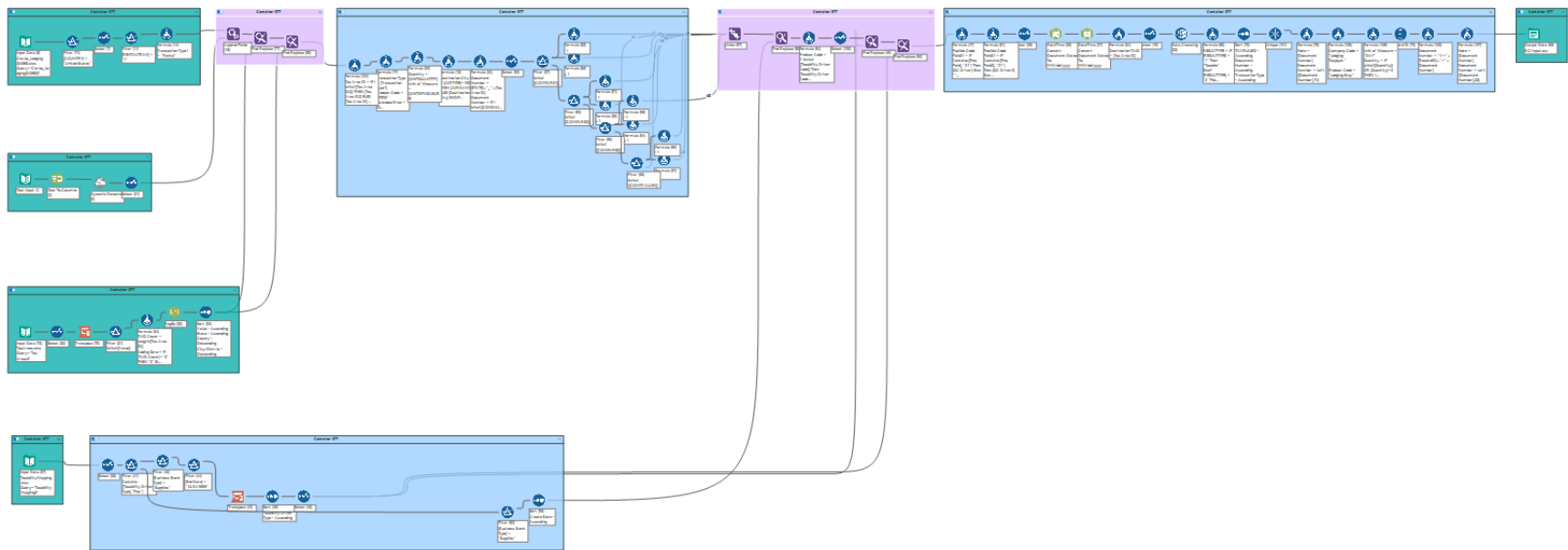
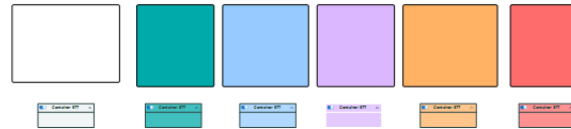
Location: C:\Documents

Workflow Owner(s): Marilyn Clarke
Business contact: Marilyn Clarke

Target audience: O Series Testers

Purpose: Generate Test Transactions that activate a list of provided rules

Version 1.0.0
1.0 - Original Workflow from 2022



All examples are anonymized or synthetic. No proprietary data is disclosed.